

SHRESTH PAL

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Education

University of Texas at Arlington

Master of Science in Computer Science

Aug 2021 – May 2023

GPA : 3.83/4.00

Jaypee Institute Of Information Technology, India

Bachelor of Technology in Computer Science

Aug 2015 – May 2019

Work Experience

Software Engineer

MS Infotech LLC, US (Remote)

Jul 2024 – Present

- Led development of an **intelligent claims processing system** for insurance clients using **Java 17** and **Spring Boot 3.0**, implementing virtual threads to handle 1000+ concurrent claims, reducing processing time by 40%
- Enhanced document analysis capabilities by integrating **Claude 3 API** with a custom **LangChain4j** pipeline that automatically extracts, classifies, and validates insurance claim information, reducing manual review time by 65%
- Optimized system performance by implementing **annotation-based smart caching** with **Spring AOP** and **Redis**, resulting in claims verification response times improving from 120ms to 35ms during peak loads.
- Streamlined data exchange between legacy systems and modern microservices using type-safe **Java Records** and automated object mapping with **MapStruct**, eliminating data conversion errors that previously affected 5% of all claims.
- **Skills:** Java, Virtual threads, Claude 3 API, LangChain4j, MapStruct, Java Records, Redis, Spring AOP

Software Engineer

UTARI, Fort Worth, Texas

Sep 2022 – Jun 2024

- Designed and developed a Cloud application for data extraction from Android Wear devices using Java REST APIs.
- Integrated data from **Fitbit and Google Health**, visualized metrics on a **ReactJS dashboard**.
- Implemented an end-to-end data pipeline using **Apache Airflow** reducing data ingestion time by 40% and ensuring timely availability of high-quality health metrics.
- Integrated machine learning models with SageMaker enabling automated model deployment and real-time monitoring.
- Employed **OAuth 2.0** for authentication, **MongoDB** for data storage, and **AWS EC2** for deployment.
- Implemented **CI/CD** with Jenkins, containerized services using **Kubernetes and Docker**.
- **Skills:** Java Spring, Android, ReactJS, MongoDB, AWS, REST, Airflow, OAuth, Kubernetes

Software Developer

NTT Data Services

July 2019 – Feb 2020

- Built and deployed enterprise web applications using **Java Spring Boot**.
- Added elastic search feature in e-commerce websites and **reduced the average search time by 50%**.
- Utilized **PostgreSQL, JDBC, and JPA** for database management.
- Developed microservices on Azure, enabling content management for **10+ enterprises**.
- Used Mockito framework and wrote **Junit** test cases and integration tests.
- Employed caching mechanisms with **Redis** to optimize performance & reduce database load, improving response times.
- Collaborated with cross-functional stakeholders to build projects using an **agile** framework.
- **Skills:** Spring Boot, Java, PostgreSQL, MVC, Elastic Search, Azure, Redis, Docker

Projects

Student Academic Portal | JavaScript, NodeJS

December 2022

- Developed a website for University of Texas using MVC model and NodeJS framework.
- Wrote APIs using REST framework and wrote service layer, business logic and created Data access Object classes.
- Wrote Unit and Integration test cases using JEST Framework and utilised Mocking Function API.
- Scheduled, updated and allotted tasks to the team mates as a Scrum Master.

ChatGenius | Microsoft BOT Framework, Sklearn, Azure

October 2022

- Designed and developed an end-to-end chat application on Azure using Microsoft BOT framework and MVC architecture, enabling users to place meal orders, while incorporating Machine Learning techniques such as Language Understanding Intelligent Service and Bing Spell Check API to enhance the BOT's intelligence.

Technical Skills

Languages: Java, JavaScript, Python, C++, Bash, HTML, CSS, SQL

Databases/Web Technologies: MySQL, MongoDB, Hadoop, DynamoDB, CosmosDB, Spark, Kafka, Redis, REST API

Frameworks/Tools: Spring Boot, J2EE, NodeJS, ReactJS, Vue.js, Flask, Jersey JAX-RS, GraphQL, JIRA, Kubernetes

Cloud Technologies & DevOPS: AWS, EC2, S3, Azure, Google Cloud Platform (GCP), Git, Docker, Jenkins